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Session: ISUW 2026

PRESENTATION TOPIC: SANJOGA: A GIS-ENABLED REAL-TIME FEASIBILITY ASSESSMENT TOOL FOR ACCELERATING NEW ELECTRICITY CONNECTION

Presented By

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INTRODUCTION

•. Electricity distribution utilities in India face increasing pressure to provide **reliable, affordable, and timely services**

•Release of **new power connections** is a highly visible and regulator-monitored function.

•**Site feasibility assessment** is the first and most critical step in the connection process.

Traditional methods rely on:
•Manual site visits
•Handwritten reports
•Fragmented information systems

These challenges lead to:
•Inconsistent decisions
•Process delays
•Revenue leakages

To address this, **TP Western Odisha Distribution Limited (TPWODL)** developed **SANJOGA**.

- **SANJOGA** is a **GIS-enabled, real-time feasibility assessment mobile application**.
- It integrates with enterprise systems such as:
  - Billing systems
  - GIS platform
  - Field inspection workflows
- Represents a significant step toward **digital transformation in distribution operations**.

Broder Steps of New Service Connection:

1. Registration/Application
2. Document verification
3. Commercial & Technical Feasibility, Site Visit
4. Meter & NW installation
5. Consumer & bill generation

CONTEXT

Growing urbanization and rural electrification have significantly increased **new connection demand**. Operational challenges in widespread DISCOMs (Having higher geographical areas)

Regulatory bodies mandate **strict timelines** for service release and monitoring of performance standards.

Customer expectations have shifted toward **faster, transparent, and digital service delivery**.

The traditional feasibility assessment process involved:

- Physical inspections without geo-tag validation
- Manual documentation and approvals
- Limited real-time coordination between departments

Lack of system integration resulted in:

- Data duplication
- Limited visibility for management
- Increased operational inefficiencies

Need identified for a **technology-driven, transparent, and integrated solution**.

This necessity led to the conceptualization of a **GIS-enabled digital feasibility platform**.

RELEVANCE

Reduced Delays

Reduces **process delays and operational inefficiencies**.

Revenue Minimization

Minimizes **revenue leakage** through accurate load assessment and documentation.

Regulatory Compliance

Ensures **regulatory compliance** with service release timelines.

Enhanced Transparency

Enhances **transparency and accountability** in field operations.

Customer Satisfaction

Directly impacts **customer satisfaction** and public perception of the utility.

Digital Transformation

Contributes to overall **digital transformation of distribution utilities**.

Aligns with the vision of **smart and technology-enabled power distribution**

PRESENTATION ON THE TOPIC (1/2)

Site Inspection Through **SANJOGA: A GIS-ENABLED REAL-TIME FEASIBILITY ASSESSMENT TOOL FOR ACCELERATING NEW ELECTRICITY CONNECTION**

- Allocation of New connection for site inspection
- Quick option to call the applicant
- Neighbouring consumers with dues within a 50-meter radius
- Nearest Distribution Network for Technical Feasibility
- User friendly Simple menu & Local Language Compatibility
- Real –Time Site Condition Update
- Offline mode for areas with poor connectivity

PRESENTATION ON THE TOPIC (2/2)

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SANJOGA App Features

Fig-1: GIS-Based Location Intelligence

Fig-2: Allocation and direct calling to the applicant

Fig-3: Neighbouring consumers with Dues for commercial feasibility

Fig-4: Nearest Distribution NW for technical feasibility

Fig-5: Local language compatibility

Fig-6: Real time Site condition update

USE CASE / CASE STUDY

- Scenario: Applicant requests a **new electricity connection** in a semi-urban area (Area of TPWODL is 48000 Sq KM)
- Field engineer receives assignment through **Sanjoga mobile application**.
- At site, engineer:
  - Captures **geo-tagged location & photographs**
  - Maps nearest pole/transformer using GIS
  - Dues verification of the plot and neighbouring consumers using GIS
  - Assesses load requirement digitally
  - Calculates distance from existing network
- Feasibility report submitted **instantly through app** (no manual paperwork).
- Data synced with the central system for approval and billing integration.

Outcome & Impact

- **Reduction in Inspection Cycle Time** significantly. Site inspection time reduced from 2-5 days to 1-2 days
- **Increased Transparency and Ethical Practices:** Geo-tagged evidence, digital timestamps, and automated workflows reduce discretionary decisions and scope for malpractice.
- Eliminated manual errors and inconsistent decision-making.
- Improved transparency and accountability.
- Faster service release → Higher customer satisfaction.
- **Management Oversight & Data Analytics**

Sanjoga generates data insights that allow leadership to:

- Identify high-load areas
- Monitor staff performance
- Understand typical causes of infeasibility
- Plan network strengthening where needed
- This aligns field operations with corporate planning.

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)